

# Quality Management Framework and Inspection Protocols for Utility-Scale Solar PV Projects

## 1. Quality Management Overview

In the execution of utility-scale Engineering, Procurement, and Construction (EPC) projects, a rigorous distinction between Quality Assurance (QA) and Quality Control (QC) forms the foundation of project integrity. Quality Assurance represents the proactive, process-oriented framework designed to prevent defects. It encompasses the systematic activities implemented within the quality management system—such as auditing, procedural development, and training—to provide confidence that the project will fulfill all quality requirements. Quality Control, conversely, is the reactive, product-oriented execution of these plans. It involves the operational techniques and activities, such as physical inspections, material testing, and commissioning verification, utilized to measure the physical characteristics of the installation against predefined acceptance criteria. In an EPC environment, QA dictates how the project team will ensure quality, while the QC team physically verifies that the subcontractor's workmanship meets the engineering design.

The Quality Plan (QP) serves as the master governing document for project quality. It establishes the organizational structure, delineates responsibilities, and defines the processes necessary to achieve the project's quality objectives in alignment with ISO 9001 and specific lender requirements, such as the IFC Performance Standards<sup>1</sup>. A comprehensive QP contains sections detailing management responsibility, document control mechanisms, procurement and material tracking, non-conformance management workflows, inspection and testing protocols, internal auditing schedules, and final handover procedures. The QA/QC Manager develops the QP during the mobilization phase. It must subsequently be approved by the EPC Project Director, the Client, and the Lender's Independent Engineer (IE) prior to the commencement of any permanent works.

Derived from the overarching Quality Plan, the Inspection and Test Plan (ITP) is a highly detailed, discipline-specific matrix that dictates the chronological sequence of quality interventions. The ITP outlines every step of the construction process, detailing the specific activity to be inspected, the reference standard or engineering drawing, the quantitative or qualitative acceptance criteria, the required quality record, and the level of intervention required by the Subcontractor, the EPC QC team, and the Client/IE. Discipline-specific QC Engineers (Civil, Mechanical, Electrical) develop the ITPs, which are then reviewed and approved by the QA/QC Manager and the Client.

To manage the flow of construction and ensure critical verifications are not bypassed, ITPs utilize specific inspection categories. A Hold Point (H) is a critical construction step where work absolutely cannot proceed without the physical presence and formal sign-off of the designated inspector. Advancing past a Hold Point without authorization constitutes a severe

procedural violation, immediately triggering a Non-Conformance Report (NCR). A Witness Point (W) requires the subcontractor to notify the inspector of an upcoming activity; however, if the inspector does not attend at the scheduled time, the work may proceed. A Review (R) is a documentary check where the inspector examines quality records, such as Mill Test Certificates (MTCs), calibration certificates, or concrete pour cards, without needing to witness the physical execution of the work. Monitor (M), or Surveillance, involves the random, ongoing observation of the work process to ensure general adherence to quality standards and safety practices, usually without a specific sign-off requirement for each occurrence.

The QA/QC Manager possesses overriding authority regarding project quality, acting independently from the construction schedule pressures. This includes the absolute authority to issue a Stop Work Order (SWO). An SWO is deployed under conditions where persistent, systemic non-conformances jeopardize structural integrity or electrical safety, when unapproved or counterfeit materials are introduced to the site, when construction teams repeatedly advance past Hold Points without authorization, or when execution procedures severely violate safety and quality protocols, risking catastrophic equipment failure or environmental contamination.

## 2. Civil QC Protocols (PLUMBLINE)

Civil quality control ensures the foundational and geotechnical integrity of the solar PV plant, dictating the physical longevity of the infrastructure against environmental degradation. Earthworks and site grading operations are heavily tested for soil bearing capacity and compaction density to ensure the subgrade can support the required structural loads without differential settlement. The primary assessment is the soil compaction test, often evaluated using the Standard Proctor test or equivalent regional methodologies such as GOST 22733<sup>3</sup>. Acceptance criteria typically demand 95% of the maximum dry density for general site grading and 98% beneath critical structural foundations, roadways, and the substation footprint. Field testing is executed utilizing calibrated nuclear density gauges or sand-cone methods, with test frequencies strictly dictated by the civil ITP.

Pile installation, whether driven galvanized steel sections or bored concrete friction piles, requires rigorous geometric and geotechnical verification. For driven tracker or fixed-tilt

foundations, QC inspectors monitor positional accuracy (typical tolerance  $\pm 50$  mm),

verticality or plumbness (typical tolerance  $\pm 1$  degree), twist, and top elevation. Inspectors strictly verify the "set" criteria—the depth achieved per hydraulic hammer blow—to confirm that the soil skin friction and end-bearing capacity match the geotechnical design models. If a pile reaches refusal before achieving the design embedment depth, engineering intervention is required, often resulting in a concession to pre-drill the location or design a concrete collar. Pile driving records are comprehensive logs that capture the coordinates, depth, driving duration, and set criteria for every single pile on the site. Rejection criteria include excessive deformation of the pile head, severe galvanizing damage, or driving out of specified spatial tolerances. Concrete works are monitored across their entire lifecycle. Prior to pouring, the shuttering (formwork) is inspected for dimensional accuracy, rigidity, and cleanliness. The reinforcement (rebar) is verified against the approved Bar Bending Schedule (BBS) for correct diameter,

spacing, lap splice lengths, and the proper placement of cover blocks to prevent corrosion. During the pour, the workability of the fresh concrete is verified via a slump test, typically targeting 100-150 mm depending on the approved mix design. QC teams cast concrete cubes or cylinders for compressive strength testing, curing them in controlled water baths. Testing is conducted at 7 days to predict early strength gain and at 28 days for final structural acceptance. The curing process itself is a monitored activity; curing logs ensure the concrete is kept moist via burlap or treated with chemical curing compounds for the specified duration (usually 7-14 days) to prevent plastic shrinkage cracking.

Internal road construction demands rigorous sub-base and base course compaction testing, utilizing Proctor and nuclear gauge methods. The California Bearing Ratio (CBR) is tested to evaluate the mechanical strength of the road subgrade, ensuring it can withstand the axle loads of heavy delivery trucks and ongoing Operations and Maintenance (O&M) vehicles. Final surface finishes, layer thicknesses, and grading slopes (camber) designed to facilitate water runoff are verified against the approved civil cross-sections.

Drainage networks and culverts are inspected for correct invert levels, trench dimensions, and proper bedding material to prevent pipe crushing. The joint integrity of HDPE or concrete pipes is verified prior to backfilling. Post-installation, the compaction of the backfill around and above culverts is a critical Hold Point to prevent subsequent road collapse. Flow tests or visual gradient checks are performed to guarantee that stormwater will gravity-drain toward designated retention basins or natural watercourses without pooling within the solar array.

The perimeter security fencing is inspected for post foundation embedment depth, the quality of the concrete footing, post verticality, chain-link or weld-mesh tension, and the proper installation of anti-burrowing features and barbed wire overhangs. Gate operation, grounding connections to the main earth grid, and integration points for security infrastructure (such as CCTV camera mounts) are also verified against the security design package.

The construction of the control room and the 33/220kV substation building involves comprehensive architectural and structural checks. QC oversees the foundation inspections, masonry or structural steel erection, and roofing weatherproofing—often utilizing water ponding tests to detect leaks. The installation of internal cable trenches, architectural finishes, HVAC systems, and fire-rated doors must strictly comply with both the structural drawings and local fire safety codes.

### **3. Mechanical / Structural QC Protocols (TORQUE)**

Mechanical quality control focuses on the structural erection and kinetic functionality of the solar array, which is particularly critical for single-axis tracking systems subjected to dynamic wind loads.

The erection of the Module Mounting Structure (MMS), including tracker torque tubes, purlins, slew drives, and cross-members, requires high-precision assembly. QC inspectors verify that all structural tolerances meet the specific manufacturer's installation manual to ensure no binding or excessive friction occurs during tracker rotation. Torque records are heavily scrutinized; every structural bolt must be tightened to the specified torque using a calibrated torque wrench. Torque marks—paint lines drawn across the bolt head, nut, and surrounding metal—are applied immediately after torquing to visually confirm completion and to serve as an

indicator for any future loosening due to vibration or thermal expansion. Any damage to the hot-dip galvanized coating sustained during handling or bolting must be repaired using a zinc-rich cold galvanizing compound conforming to ASTM A780, ensuring the dry film

thickness exceeds  $90 \mu\text{m}$ <sup>5</sup>.

During module mounting and installation, frames are inspected for proper alignment, exact clamp positioning according to the manufacturer's permitted clamping zones, and the application of the correct clamp torque. Visual inspections screen for scratched front glass, bent aluminum frames, or damaged polymer backsheets. Module handling requirements are strictly enforced; carrying modules by their junction box cables, stacking them improperly without protective corners, or resting them on sharp structural edges is strictly forbidden. Any damage, such as deep backsheet scratches exposing inner layers or shattered glass, is deemed rejectable, and the module must be quarantined.

To detect non-visible damage, such as microcracks induced during transit, Electroluminescence (EL) testing is employed on a sampling basis. Transportation vibration profiles and shock tests are initially governed by standards such as IEC 62759-1 to ensure packaging integrity<sup>7</sup>. Once on site, EL testing involves applying a forward bias DC current to the module in a dark environment, causing the cells to emit infrared light captured by a specialized camera. Inactive cell areas, severe dendritic microcracks, or soldering defects appear dark in the resulting image<sup>9</sup>. Modules exhibiting cracks that propagate across multiple busbars or that show significant inactive areas are rejected, often triggering a vendor warranty or transit insurance claim.

Tracker commissioning checks verify the operational limits and software algorithms of the dynamic system. Mechanical QC verifies motor alignment, Network Control Unit (NCU) communications, and the physical actuation of limit switches. The backtracking algorithm is tested to ensure that the trackers adjust their angles to prevent row-to-row shading during low-sun periods in the early morning and late evening<sup>11</sup>. Finally, the wind stow position is physically verified, ensuring the tracker can rapidly rotate to its designated safe angle (often near 0 degrees or a high tilt, depending on the aerodynamic profile) during high wind speed events, preventing catastrophic torsional failure.

## **4. Electrical QC Protocols (OHMMETER)**

Electrical quality control guarantees the energy conversion efficiency, precise metering, and personnel safety of the entire power generation and distribution network.

DC cable installation involves visual inspection for jacket integrity, correct UV resistance rating, and proper color-coding or tagging. Cable routing must respect the manufacturer's minimum bending radii to prevent localized stress on the cross-linked polyethylene (XLPE) insulation.

Cable trays must not exceed specified fill ratios to allow for adequate heat dissipation, and neat wire management (using UV-rated zip ties or stainless-steel clips) is required to prevent cables from hanging loose or rubbing against sharp MMS edges. DC cable terminations (MC4 or equivalent) are inspected for complete mating (indicated by an audible click), the correct crimping depth using the specified tool, and the absence of dirt or moisture prior to connection. Insulation Resistance (IR) is tested using a megohmmeter. For DC systems

operating above 120V, IEC 62446-1 requires an IR value greater than or equal to 1 M $\Omega$ , measured at 1000V DC for 60 seconds<sup>13</sup>. Strict polarity checks are mandatory before connecting strings to the combiner box or inverter to prevent catastrophic short circuits and diode destruction.

Pre-commissioning string testing complies strictly with IEC 62446-1<sup>14</sup>. The open-circuit voltage (Voc) and short-circuit current (Isc) of each string are measured. Acceptance criteria require the measured Voc to be within  $\pm 5\%$  of the expected value (adjusted for the actual module temperature) and the Isc to be within  $\pm 10\%$  of the expected value (adjusted for the actual plane-of-array irradiance)<sup>13</sup>. For deeper diagnostic insight, I-V curve tracing is utilized to capture the full electrical characteristic of the string. Anomalies in the curve shape provide specific defect indications: steps or plateaus indicate partial shading or bypass diode activation; a lower Isc than expected indicates heavy soiling or severe uniform degradation; and a lower Voc than expected indicates module mismatch, shorted bypass diodes, or a missing module in the series string<sup>13</sup>.

Earthing continuity testing verifies the integrity of the grounding network using the Fall of Potential (Three-Point) method in accordance with IEEE 81<sup>15</sup>. The true earth resistance is obtained by placing the potential probe at precisely 62% of the distance between the earth electrode and the current probe, targeting the zero-potential plateau<sup>16</sup>. The DC system structural earthing continuity is tested using a micro-ohmmeter to ensure a low-resistance path (typically  $< 1 \Omega$ ) exists across all MMS joints, motor housings, and inverter chassis back to the main earth grid, protecting personnel from step and touch potentials.

Medium Voltage (33kV) cable installation requires meticulous inspection during the laying process to verify proper trench depth, the installation of sand bedding, warning tape, and mechanical protection tiles. Terminations and straight-through joints are critical Hold Points, requiring verified clean environments and certified jointers. MV cables undergo Very Low Frequency (VLF) testing (0.1 Hz) rather than traditional DC high-pot testing, as DC testing can inject localized space charges that permanently damage XLPE insulation. Per IEEE 400.2, a 33kV cable is typically subjected to a VLF sinusoidal test voltage of  $3U_0$  (approximately 57kV peak) for a duration of 15 to 30 minutes, during which the insulation must demonstrate zero dielectric breakdown<sup>17</sup>.

Substation transformer testing follows IEC 60076. QC witnesses pre-commissioning tests including Winding Resistance, Insulation Resistance, Turns Ratio (TTR), Vector Group verification, and Sweep Frequency Response Analysis (SFRA). SFRA detects internal core or winding deformation resulting from transit shocks or short-circuit forces by analyzing the electrical transfer function across a wide frequency range<sup>20</sup>. Additionally, Dissolved Gas Analysis (DGA) is performed on oil samples. Methods such as the Duval Triangle (IEC 60599) are used to plot the ratios of dissolved hydrocarbon gases (methane, ethylene, acetylene) to accurately diagnose incipient thermal faults (T1, T2, T3) or electrical arcing, separating these critical issues from benign oil oxidation<sup>22</sup>.

Switchgear testing is executed per IEC 62271-100. QC verifies the vacuum circuit breaker's contact resistance using a micro-ohmmeter, ensuring the measured value does not exceed 1.5 times the manufacturer's factory test value<sup>25</sup>. Timing tests confirm the precise opening and closing speeds of the breaker poles. Protection Current Transformers (CTs) within the switchgear are tested per IEC 61869-2 to verify accuracy classes (e.g., 0.2S for precision revenue metering, 5P20 for protection). The knee point voltage is verified to ensure the CT core will not magnetically saturate during maximum fault currents, which would blind the protection relays<sup>27</sup>. Finally, protection relays undergo secondary injection testing to verify that their trip logic and timing coordinate perfectly with the power system study.

## 5. Material Inspection & MTC Verification

Material quality is validated long before physical incorporation into the site works through strict supply chain controls and meticulous receiving inspections.

A Mill Test Certificate (MTC) is a paramount quality assurance document generated by the raw material manufacturer. Governed by standard EN 10204, a Type 3.1 MTC is issued by the manufacturer's independent quality department, certifying that the chemical composition and mechanical properties—such as yield strength, ultimate tensile strength, and Charpy impact values—meet the specified structural grade. A Type 3.2 MTC demands an additional layer of scrutiny, including verification and counter-signature by an independent Third-Party Inspector (TPI)<sup>30</sup>. The QC engineer visually verifies that the unique heat numbers physically stamped or stenciled on the delivered steel members perfectly match the data on the MTC.

Third-Party Inspection (TPI) involves contracting an independent agency (e.g., SGS, TÜV, Bureau Veritas) to witness Factory Acceptance Tests (FAT) at the manufacturer's facility. TPIs are mandated for high-value, critical equipment such as main power transformers, 33kV switchgear, and PV modules. The TPI verifies performance metrics against the approved datasheet and issues an inspection release note before the equipment is permitted to ship to the site.

The incoming material inspection process at the site follows a strict step-by-step protocol:

1. **Unloading:** Materials are safely unloaded and staged in a designated receiving quarantine area.
2. **Document Verification:** The QC inspector checks the packing list, Bill of Lading, and corresponding MTCs against the original Purchase Order.
3. **Visual & Dimensional Inspection:** Materials are inspected for transit damage, correct dimensional tolerances, coating integrity (e.g., measuring galvanizing thickness using a dry film thickness gauge), and precise quantities.
4. **MIR Generation:** A Material Inspection Request (MIR) is formally raised to the Client/IE, attaching all supporting evidence, photographs, and MTCs.
5. **Disposition:** Upon Client/IE approval, the materials are physically moved to the "Accepted" laydown yard and released for construction.

Materials failing the inspection process are immediately tagged with high-visibility "REJECTED" or "QUARANTINE" labels. They are physically segregated into a restricted, barricaded quarantine zone to absolutely prevent their accidental incorporation into the works. An NCR is raised against the supplier, initiating the return, replacement, or concession process.

## 6. Non-Conformance Management

A Non-Conformance Report (NCR) is a formal control document raised when a material, installation process, or final product deviates from the project specifications, approved IFC drawings, or applicable codes. NCRs can be raised by the EPC QC team against a subcontractor's work, or by the Client/IE against the EPC contractor.

A standard NCR contains comprehensive data fields: a unique tracking number, the date of occurrence, specific location coordinates, the exact reference documents violated (e.g., drawing number or specification clause), a detailed factual description of the deviation, photographic evidence, a Root Cause Analysis (RCA) utilizing methods like the "5 Whys," a proposed corrective action to fix the immediate issue, a preventive action to stop recurrence, and formal sign-off blocks for the engineering, quality, and client representatives.

Upon raising an NCR, five primary disposition options are evaluated:

1. **Use As Is:** The deviation is evaluated by engineering and determined not to affect the structural form, fit, or electrical function. This requires a formal concession and sign-off from the Engineer of Record and the Client.
2. **Rework:** The item is brought fully back into compliance with the original specifications without leaving any trace of the defect (e.g., extracting and re-driving a shallow pile, or re-torquing a loose structural bolt).
3. **Repair:** The item is fixed to be functionally safe and structurally sound, though it may not perfectly match the original aesthetic or fabrication specification (e.g., welding a reinforced patch plate onto a misaligned steel beam). This always requires specialized engineering approval.
4. **Reject/Replace:** The item cannot be safely salvaged and must be entirely removed from the site and replaced with a new, compliant unit.
5. **Scrap:** The material is entirely written off, destroyed, or recycled to prevent accidental reuse.

The NCR close-out process requires the QC inspector to physically verify the executed repair or rework. The NCR document is updated with closing photographs, re-inspection reports (e.g., a passing test result after replacing a faulty cable), and signatures from the EPC QA/QC Manager and the Client/IE, officially closing the issue.

The NCR Register is a live, dynamic tracking matrix (maintained in Excel or specialized QMS software) detailing the NCR number, date raised, brief description, location, responsible party, current status (Open/Closed), and aging (days open). This register is reviewed in weekly quality meetings to identify systemic trends.

**Table: 20 Common Solar EPC NCRs and Typical Dispositions**

| # | Typical Non-Conformance Event                                          | Typical Disposition & Action                                          |
|---|------------------------------------------------------------------------|-----------------------------------------------------------------------|
| 1 | Driven pile reaches depth refusal before design embedment due to rock. | <b>Repair:</b> Pre-drill the location or design a concrete encasement |

|    |                                                                     |                                                                                                     |
|----|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
|    |                                                                     | collar.                                                                                             |
| 2  | Pile verticality exceeds the specified 1-degree tolerance.          | <b>Rework:</b> Extract the pile and re-drive to within tolerance.                                   |
| 3  | Concrete slump test fails upon delivery (too dry or too wet).       | <b>Reject:</b> Reject the transit mixer; do not permit pouring.                                     |
| 4  | Concrete 28-day cube strength fails to meet the 30 MPa requirement. | <b>Repair:</b> Conduct non-destructive core testing; if structurally viable, accept via concession. |
| 5  | Structural galvanizing deeply scratched during mechanical erection. | <b>Rework:</b> Clean surface and repair with cold zinc paint per ASTM A780.                         |
| 6  | Structural bolt found under-torqued during surveillance.            | <b>Rework:</b> Re-torque to specification and apply torque indicator marks.                         |
| 7  | Solar module glass shattered during manual handling.                | <b>Reject/Replace:</b> Remove the module from the site immediately.                                 |
| 8  | Module backsheet scratched, exposing inner conductive layers.       | <b>Reject/Replace:</b> Quarantine the module; do not install to prevent ground faults.              |
| 9  | EL test reveals severe dendritic microcracks post-transit.          | <b>Reject/Replace:</b> File a transit insurance claim using shock sensor data.                      |
| 10 | DC cables installed with a bend radius tighter than 8x OD.          | <b>Rework:</b> Re-route and re-secure the cable to relieve insulation stress.                       |

|    |                                                                         |                                                                                                    |
|----|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| 11 | MC4 connector crimped utilizing an unapproved, generic tool.            | <b>Reject/Replace:</b> Cut off the connector, discard, and re-crimp with the OEM tool.             |
| 12 | DC string polarity reversed at the combiner box termination.            | <b>Rework:</b> Disconnect, verify polarity with a multimeter, and re-terminate correctly.          |
| 13 | String Insulation Resistance (Megger) fails, measuring < 1 M $\Omega$ . | <b>Repair:</b> Sectionalize the string, locate the ground fault, and replace the damaged wire.     |
| 14 | IV Curve trace displays a distinct bypass diode step/plateau.           | <b>Reject/Replace:</b> Identify and replace the defective module within the series string.         |
| 15 | Substation earth grid Fall of Potential measures > 1 $\Omega$ .         | <b>Repair:</b> Drive additional deep earth rods and exothermic weld to the grid.                   |
| 16 | 33kV MV cable VLF test fails due to dielectric breakdown.               | <b>Reject/Replace:</b> Cut out the faulted section and install an approved straight-through joint. |
| 17 | Transformer oil DGA shows high acetylene levels (T3/Arcing fault).      | <b>Reject:</b> Do not energize; return to the manufacturer for active core repair.                 |
| 18 | MV Switchgear vacuum breaker contact resistance > 1.5x limits.          | <b>Rework:</b> Rack out the breaker, clean the tulip contacts, apply contact grease, and re-test.  |
| 19 | Protection CT polarity inverted during secondary                        | <b>Rework:</b> Reverse the secondary wiring connections at the terminal                            |

|    |                                                                         |                                                                                                  |
|----|-------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
|    | wiring.                                                                 | block.                                                                                           |
| 20 | Subcontractor backfills an underground cable trench without QC witness. | <b>Rework:</b> Re-excavate the trench to permit mandatory visual inspection of the sand bedding. |

## 7. Punch List Management

A punch list (or snag list) is a comprehensive inventory of outstanding, incomplete, or defective items identified during the final joint walkthroughs conducted by the EPC QC team and the Client's IE just prior to Mechanical Completion or Provisional Acceptance.

Punches are strictly categorized to manage handover risk:

- **Category A (Major):** Critical defects that prevent the safe energization, operability, or mechanical completion of the system. Examples include open earth grid connections, missing switchgear protection relays, or exposed live high-voltage conductors. Every single Category A punch must be completely closed and signed off before the system can be legally turned over or energized.
- **Category B (Minor):** Minor, non-critical, or cosmetic defects that do not impair operation, performance, or safety. Examples include missing equipment identification labels, scratched paint requiring touch-up, or general site clean-up. Category B punches can be deferred until after Provisional Acceptance, usually with a financial retention held by the client to guarantee their eventual closure.

The close-out process requires the construction team to systematically rectify the listed items. The QC engineer photographs the correction and compiles a "Punch List Close-out Report." The Client/IE reviews this documentary evidence or conducts a localized re-walkdown. If satisfied, the IE signs off on the specific line item. The ultimate authority to accept a punch close-out and issue the completion certificate rests exclusively with the Client's Representative or the Independent Engineer.

## 8. Handover Documentation Package (DOSSIER)

The Quality Dossier is the final, exhaustively indexed compilation of all quality records generated throughout the project lifecycle. It provides the Client, Lenders, and the O&M team with absolute traceability and assurance that the plant was built precisely to specification. The package is organized hierarchically by discipline (Civil, Mechanical, Electrical) and further subdivided by facility (Solar Field blocks, Substation, Transmission Line). A complete dossier contains:

1. The finalized Project Quality Plan and all approved ITPs.
2. All Material Inspection Reports (MIR) supported by EN 10204 3.1/3.2 MTCs.
3. Geotechnical survey reports and comprehensive pile driving logs.
4. Concrete pour cards, slump logs, and 28-day cube crush test results.

5. Structural torque logs and tracker calibration certificates.
6. Module flash test data (factory) and site EL testing reports.
7. Electrical pre-commissioning reports (Megger, IV Curve, VLF, Transformer routine tests, CT/PT tests).
8. The closed NCR register and the signed-off Punch List.
9. Calibration certificates for all test equipment utilized during the project.

The Lender's Engineer reviews the dossier to ensure no critical Category A punches remain, all NCRs are formally closed with proper engineering justification, and test results meet contractual guarantees. They specifically audit the traceability—ensuring that if a test failed, there is a clear, linked re-test record proving eventual compliance. Furthermore, performance metrics such as the Performance Ratio (PR) are scrutinized. To prove the plant's true electrical health to lenders, a temperature-corrected PR metric (calculated per IEC 61724-1) is utilized to normalize thermal losses resulting from high ambient temperatures, providing a highly accurate baseline for O&M<sup>32</sup>.

As-Built documents are a critical handover requirement. They reflect the exact, final dimensions, routing, and parameters of the installed plant, superseding the "Issued for Construction" (IFC) drawings. The construction team redlines the IFC drawings in the field to show deviations (e.g., shifting a cable trench to avoid unforeseen bedrock). The EPC Engineering department then updates the CAD files and issues the formal, stamped As-Built package.

The O&M Manual, compiled by the EPC, provides the long-term operators with the information necessary to safely run the plant. It covers system start-up/shutdown sequences, LOTO (Lockout/Tagout) safety protocols, preventive maintenance schedules, troubleshooting guides, warranty certificates, and specific OEM equipment manuals for inverters, trackers, and switchgear.

## 9. Real Document Templates

### Quality Plan Structure (Table of Contents Level)

| Section | Title                         | Description of Contents                                                                                              |
|---------|-------------------------------|----------------------------------------------------------------------------------------------------------------------|
| 1.0     | Scope and Purpose             | Defines project boundaries, applicable codes (IEC, ISO), and overarching quality goals for the HELIOS 100MW project. |
| 2.0     | Quality Policy and Objectives | Management commitment, KPIs (e.g., <1% NCR rate), and continuous improvement strategies.                             |

|             |                                         |                                                                                                       |
|-------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>3.0</b>  | Project Organization & Responsibilities | Organogram detailing authority of QA/QC Manager, discipline QC Engineers, and interfaces with the IE. |
| <b>4.0</b>  | Document and Record Control             | Protocols for controlling IFC drawings, redlines, and maintaining the central Quality Dossier.        |
| <b>5.0</b>  | Procurement and Material Control        | Procedures for supplier evaluation, executing FATs, verifying MTCs, and generating MIRs.              |
| <b>6.0</b>  | Construction Quality Control            | Integration of ITPs, execution of Method Statements, and managing Hold/Witness points.                |
| <b>7.0</b>  | Non-Conformance & Corrective Action     | Workflow for raising, dispositioning, and closing NCRs, including Root Cause Analysis.                |
| <b>8.0</b>  | Inspection, Measuring, and Test Eqpt    | Calibration schedules and traceability for torque wrenches, megohmmeters, and nuclear density gauges. |
| <b>9.0</b>  | Pre-Commissioning and Handover          | Managing the Punch List, final walkthroughs, and compiling the As-Built and Dossier packages.         |
| <b>10.0</b> | Internal Quality Audits                 | Schedule and checklist for EPC QA audits against subcontractors to ensure process compliance.         |

## ITP for Civil Works (Example: Concrete Foundations)

| Activity Description | Subcontractor | EPC QC      | Client/IE   | Ref. Standard / Drawing | Acceptance Criteria                               | Quality Record      |
|----------------------|---------------|-------------|-------------|-------------------------|---------------------------------------------------|---------------------|
| Subgrade Compaction  | Execute       | Witness (W) | Review (R)  | GOST 22733 / Spec       | 98% Max Dry Density                               | NDT Density Report  |
| Formwork & Rebar     | Execute       | Hold (H)    | Witness (W) | IFC Dwg CIV-01          | Rebar spacing $\pm 10\text{mm}$ ; Cover 50mm      | Pre-Pour Insp. Card |
| Concrete Slump Test  | Execute       | Witness (W) | Monitor (M) | Project Spec            | Slump 100mm $\pm 25\text{mm}$                     | Slump Test Log      |
| Concrete Pouring     | Execute       | Monitor (M) | Monitor (M) | Method Statement        | Max drop height 1.5m, continuous vibration        | Pour Card           |
| Cube Casting         | Execute       | Witness (W) | Monitor (M) | Local Code/Spec         | Cast 6 cubes per 50m <sup>3</sup> continuous pour | Cube Register       |
| 28-Day Strength Test | Execute       | Hold (H)    | Review (R)  | Project Spec            | Compressive strength                              | Lab Test Report     |

|  |  |  |  |  |                                |  |
|--|--|--|--|--|--------------------------------|--|
|  |  |  |  |  | $\geq 30$<br>MPa at 28<br>days |  |
|--|--|--|--|--|--------------------------------|--|

### ITP for Tracker / MMS Erection

| Activity Description | Subcontractor | EPC QC      | Client/IE   | Ref. Standard / Drawing | Acceptance Criteria                                                    | Quality Record     |
|----------------------|---------------|-------------|-------------|-------------------------|------------------------------------------------------------------------|--------------------|
| Pile Driving         | Execute       | Witness (W) | Monitor (M) | IFC Dwg MEC-02          | Elev $\pm 10\text{mm}$ ,<br>Vert $<1^\circ$ ,<br>Pos $\pm 50\text{mm}$ | Pile Driving Log   |
| Torque Tube Mounting | Execute       | Witness (W) | Monitor (M) | OEM Manual              | Bearing alignment verified, frictionless rotation                      | Mech. Insp. Report |
| Structural Torquing  | Execute       | Hold (H)    | Witness (W) | OEM Manual              | M12 Bolts at 85 Nm; torque marks clearly applied                       | Torque Log         |
| Galvanizing Repair   | Execute       | Witness (W) | Review (R)  | ASTM A780               | Zinc paint dry film thickness $> 90 \mu\text{m}$                       | Insp. Report       |

|                       |         |          |             |               |                                                         |                      |
|-----------------------|---------|----------|-------------|---------------|---------------------------------------------------------|----------------------|
| Tracker Movement Test | Execute | Hold (H) | Witness (W) | T&C Procedure | Smooth rotation +55° to -55°; Stow mechanism functional | Commissioning Report |
|-----------------------|---------|----------|-------------|---------------|---------------------------------------------------------|----------------------|

### ITP for DC Electrical Works

| Activity Description  | Subcontractor | EPC QC      | Client/IE   | Ref. Standard / Drawing | Acceptance Criteria                               | Quality Record       |
|-----------------------|---------------|-------------|-------------|-------------------------|---------------------------------------------------|----------------------|
| DC Cable Routing      | Execute       | Monitor (M) | Monitor (M) | IFC Dwg ELE-03          | Min bend radius 8x OD; cable tray fill < 40%      | Routing Insp. Report |
| MC4 Crimping          | Execute       | Witness (W) | Monitor (M) | OEM Manual              | OEM die utilized, audible click, pull test passed | Insp. Report         |
| String Polarity Check | Execute       | Hold (H)    | Witness (W) | IEC 62446-1             | Correct positive/negative landing at Combiner Box | Polarity Test Report |
| Insulation Resistance | Execute       | Hold (H)    | Witness (W) | IEC 62446-1             | > 1 MΩ                                            | IR Test Sheet        |

|                             |         |          |                |                |                                                                                           |                    |
|-----------------------------|---------|----------|----------------|----------------|-------------------------------------------------------------------------------------------|--------------------|
| e                           |         |          |                |                | measure<br>d at<br>1000V<br>DC for 60<br>seconds                                          |                    |
| Voc / Isc<br>String<br>Test | Execute | Hold (H) | Witness<br>(W) | IEC<br>62446-1 | Voc $\pm$<br>5%, Isc<br>$\pm$ 10% of<br>temperat<br>ure-adjus<br>ted<br>expected<br>value | String<br>Data Log |

### ITP for Substation E&M (Example: 33kV Switchgear)

| Activity<br>Descripti<br>on | Subcontr<br>actor | EPC QC         | Client/IE      | Ref.<br>Standard<br>/<br>Drawing | Accepta<br>nce<br>Criteria                                      | Quality<br>Record       |
|-----------------------------|-------------------|----------------|----------------|----------------------------------|-----------------------------------------------------------------|-------------------------|
| Panel<br>Alignmen<br>t      | Execute           | Witness<br>(W) | Review<br>(R)  | IFC Dwg<br>SUB-01                | Plumb<br>and level,<br>securely<br>bolted to<br>base<br>channel | Mech<br>Insp.<br>Report |
| Contact<br>Resistanc<br>e   | Execute           | Hold (H)       | Witness<br>(W) | IEC<br>62271-10<br>0             | Main<br>contacts<br>< 1.5x FAT<br>value (<br>$\mu\Omega$ )      | CRM Test<br>Report      |
| Protectio<br>n CT           | Execute           | Hold (H)       | Witness<br>(W) | IEC<br>61869-2                   | Ratio<br>correct,<br>Class                                      | CT Test<br>Report       |

|                            |         |          |                |                          |                                                                                               |                         |
|----------------------------|---------|----------|----------------|--------------------------|-----------------------------------------------------------------------------------------------|-------------------------|
| Testing                    |         |          |                |                          | 5P20,<br>Knee<br>point ><br>design<br>minimum                                                 |                         |
| Relay<br>Secondary<br>Inj. | Execute | Hold (H) | Hold (H)       | Relay<br>Coord.<br>Study | Trip<br>timing<br>matches<br>curves<br><br>within $\pm$<br>5%<br>tolerance                    | Relay<br>Test<br>Report |
| VLF MV<br>Cable<br>Test    | Execute | Hold (H) | Witness<br>(W) | IEEE<br>400.2            | <b><math>3U_0</math></b><br>(57kV<br>peak) for<br>30 mins;<br>zero<br>dielectric<br>breakdown | VLF Test<br>Report      |

## NCR Form Template

**Non-Conformance Report (NCR)** **NCR No:** HELIOS-NCR-ELE-042 **Date:** 2026-10-03

**Originator:** OHMMETER (Electrical QC Engineer) **Location:** Substation Bay 2, 33kV Incomer Cable Trench **Reference Docs:** IEEE 400.2, Project Spec Sec 16.4, IFC ELE-10 **Description of Non-Conformance:** During VLF testing of the 33kV Phase A cable, a catastrophic dielectric

breakdown occurred at 18 minutes into the 30-minute  $3U_0$  test, failing the acceptance criteria. **Root Cause Analysis:** Subcontractor damaged the XLPE insulation jacket by dragging the cable across a sharp concrete edge during pulling, failing to use the mandated cable rollers. **Proposed Corrective Action:** Excavate the duct bank, cut out the damaged section of the cable, install an approved straight-through joint, and re-test the entire cable run with VLF. **Proposed Preventive Action:** Mandate the use of cable rollers and dynamometer tension monitoring during all future MV cable pulls. Toolbox talk on cable handling. **Disposition:** ☐ Use As Is ☐ Rework ☒ Repair ☐ Reject ☐ Scrap **Signatures:** [QC Inspector] [Subcontractor Rep] [Client IE - Disposition Approval] **Close-Out Verification:** Repair completed and VLF re-test passed on 2026-10-08. [QC Manager Signature]

Inspection Request (IR) Form Template

Inspection Request (IR)IR No: HELIOS-IR-CIV-105 Date Submitted: 2026-10-03 Requested Insp. Date/Time: 2026-10-04, 09:00 AM Discipline: Civil Location: Inverter Pad B4 Activity to be Inspected: Rebar and formwork prior to concrete pour.Reference ITP & Step: ITP-CIV-002, Step 3.2 Subcontractor QC Sign-off: "I certify the works are complete and ready for inspection." [Signature]EPC QC Results: [X] Approved [ ] Approved with Comments [ ] Rejected Comments: Rebar cover blocks adjusted on the North side. Ready for pour.Signatures: [EPC QC Inspector] [Client IE]

Material Inspection Report (MIR) Template

Material Inspection Report (MIR)MIR No: HELIOS-MIR-MEC-012 Date: 2026-10-03 Supplier: TrackerCorp Ltd.Material Description: Torque Tubes (150 units), Slew Drives (20 units) PO / Delivery Note No: PO-8839 / DN-4401 Inspection Checklist:

- [X] Quantities match packing list.
- [X] No visible transit damage (scratches, dents).
- [X] Galvanizing thickness tested (>90  $\mu$  m).
- [X] EN 10204 3.1 MTCs provided and heat numbers match physical stamps.

Status: [X] ACCEPTED [ ] QUARANTINED Comments: Materials cleared for transfer to Block C laydown.Signatures: [EPC QC Inspector] [Client IE]

Punch List Template

| Punch ID | Date Raised | Category (A/B) | Location      | Description of Defect             | Assigned To | Target Close Date | Status | IE Sign-off |
|----------|-------------|----------------|---------------|-----------------------------------|-------------|-------------------|--------|-------------|
| PL-E-001 | 2026-10-01  | A              | String Box 12 | Reverse polarity on String 4.     | Elec Sub    | 2026-10-02        | CLOSED | [Signature] |
| PL-M-002 | 2026-10-02  | B              | Row 44        | Missing torque mark on slew drive | Mech Sub    | 2026-10-05        | OPEN   |             |

|          |            |   |             |                                               |           |            |      |  |
|----------|------------|---|-------------|-----------------------------------------------|-----------|------------|------|--|
|          |            |   |             | bolt.                                         |           |            |      |  |
| PL-C-003 | 2026-10-03 | B | Substation  | Paint touch-up required on control room door. | Civil Sub | 2026-10-10 | OPEN |  |
| PL-E-004 | 2026-10-03 | A | Transformer | DGA sample port valve weeping oil.            | EPC Comm  | 2026-10-04 | OPEN |  |

## Handover Document Checklist

**Facility:** 100MW Solar Array & 33kV Collection System

- [ ] **1.0 Quality Management:** Final Approved Quality Plan, complete ITP index.
- [ ] **2.0 Engineering:** As-Built Drawings (Civil, Mechanical, Electrical, SCADA redlines).
- [ ] **3.0 Civil Records:** Proctor tests, pile driving logs, concrete cube test register.
- [ ] **4.0 Mechanical Records:** Structural torque logs, tracker commissioning certificates, EL imaging reports.
- [ ] **5.0 Electrical Records:** String test logs (Voc/Isc), IR Megger sheets, VLF cable test reports, Earthing FOP test reports.
- [ ] **6.0 Material Approvals:** All approved MIRs, TPI release notes, EN 10204 3.1/3.2 MTCs.
- [ ] **7.0 Non-Conformance:** Complete NCR Register (all items closed with evidence).
- [ ] **8.0 Handover:** Signed Category A Punch List, O&M Manuals, OEM Warranty Certificates.

## 10. Quality Interfaces

The QC function does not operate in a vacuum; it sits at the critical intersection of engineering, procurement, construction, and client relations.

**With Engineering:** QC relies entirely on the Engineering department's Approved for Construction (AFC) drawings and specifications to define quantitative acceptance criteria. QC inspectors are not authorized to independently accept deviations from the design. If field conditions dictate that a tolerance cannot be met (e.g., hitting immovable bedrock during piling), QC must interface with Engineering to issue a Technical Query (TQ) or Field Design

Change (FDC). Engineering then runs the calculations to adjust the criteria, which QC subsequently enforces.

**With Construction:** The relationship is inherently frictional but operationally necessary. Hold points mandate that construction crews must stop production and wait for QC verification. This requires strict scheduling and a formal 24-hour Inspection Request (IR) process. Practical management involves QC inspectors shadowing construction crews during critical phases to expedite sign-offs without causing expensive equipment downtime, balancing the need for speed with the absolute requirement for compliance.

**With Supply Chain Management (SCM):** QC interfaces heavily with SCM to review vendor quality plans, witness FATs, and conduct incoming material inspections. QC acts as the gatekeeper; procurement is typically prohibited from releasing final payments to vendors until the materials have successfully cleared the MIR process and Type 3.1/3.2 MTCs are definitively verified against the delivered physical goods.

**With Testing & Commissioning (T&C):** QC is responsible for the static, dead-system pre-commissioning phase (e.g., Megger testing, VLF, continuity checks, structural torque). Once QC formally signs off on the mechanical completion of a subsystem, demonstrating it is built to design and electrically safe, the system is handed over to the T&C team for dynamic, live-system testing (e.g., inverter synchronization, protection relay trip testing, SCADA integration).

**With Client / Independent Engineer (IE):** The IE audits the EPC's quality records to ensure the plant meets all lender requirements, including the rigorous IFC Performance Standards regarding environmental and social EHS controls<sup>1</sup>. The IE will review the NCR register, witness critical Hold points, and evaluate the final performance metrics. For example, they will audit the Performance Ratio (PR) calculation, ensuring it utilizes the temperature-corrected methodology specified in IEC 61724-1, normalizing thermal losses to prove the plant's true electrical health and revenue-generating capability to the lenders<sup>32</sup>.

## 11. Common Quality Issues on Solar Projects

Navigating quality issues requires a blend of technical expertise and procedural discipline. The following are typical scenarios encountered on utility-scale projects and their standard resolutions:

### 1. Modules received with microcracks:

- *Detection:* Discovered during sample EL testing upon arrival or post-installation<sup>9</sup>.
- *Disposition:* Reject and replace heavily damaged modules.
- *Vendor Claim Process:* The EPC initiates a transit or manufacturing warranty claim using the EL imagery combined with G-force logs extracted from shipping sensors. This claim references the IEC 62759-1 standard, which dictates acceptable transportation vibration profiles, proving the damage occurred during transit or manufacturing, not during site handling<sup>7</sup>.

### 2. Pile driving outside tolerance:

- *Issue:* A pile reaches refusal early due to rocky soil, or twists out of coordinate limits during driving.
- *Options:* The primary option is to extract and re-drive the pile. If rock is present,

engineering may approve a repair methodology involving pre-drilling the hole and grouting the pile (concrete encasement), or modifying the tracker superstructure (e.g., using specialized brackets) to absorb the positional tolerance without inducing stress on the torque tube.

**3. Concrete cube test fails at 28 days:**

- *QC Response:* Immediately issue an NCR. Construction is strictly prohibited from loading the foundation with steel structures or transformers.
- *Resolution:* The QC team performs non-destructive testing (such as a Schmidt rebound hammer test) or extracts a cylindrical core from the poured foundation for independent laboratory crushing. If the core still fails, the foundation must be demolished and recast. Alternatively, engineering can run a structural calculation to determine if the achieved lower strength is mathematically acceptable for the specific load (resulting in a "Use As Is" concession).

**4. Subcontractor re-works NCR without QC witness:**

- *Issue:* A subcontractor backfills a buried cable trench or paints over a structural repair without notifying the QC inspector to witness the Hold Point.
- *Action:* The work is categorically rejected. The subcontractor must undo the work—excavating the trench by hand to expose the sand bedding, or stripping the paint to reveal the weld—to allow the mandatory visual verification. The subcontractor bears all associated financial costs and schedule delays for this procedural breach.

**5. Lender's engineer raises a quality concern that QC had already closed:**

- *Issue:* The IE disputes an accepted deviation, such as an accepted cable bend radius or a specific test parameter.
- *Handling:* The QA/QC Manager formally escalates the issue by presenting the specific standard code (e.g., IEC, IEEE) or the manufacturer's approved installation manual that technically justifies the EPC's closure of the issue. However, if the IE's concern is based on a higher standard explicitly defined in the lender's technical requirements, the EPC must comply, re-open the NCR, and execute further corrective action to satisfy the lender.

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